Digital Systems Design (BECE102L)

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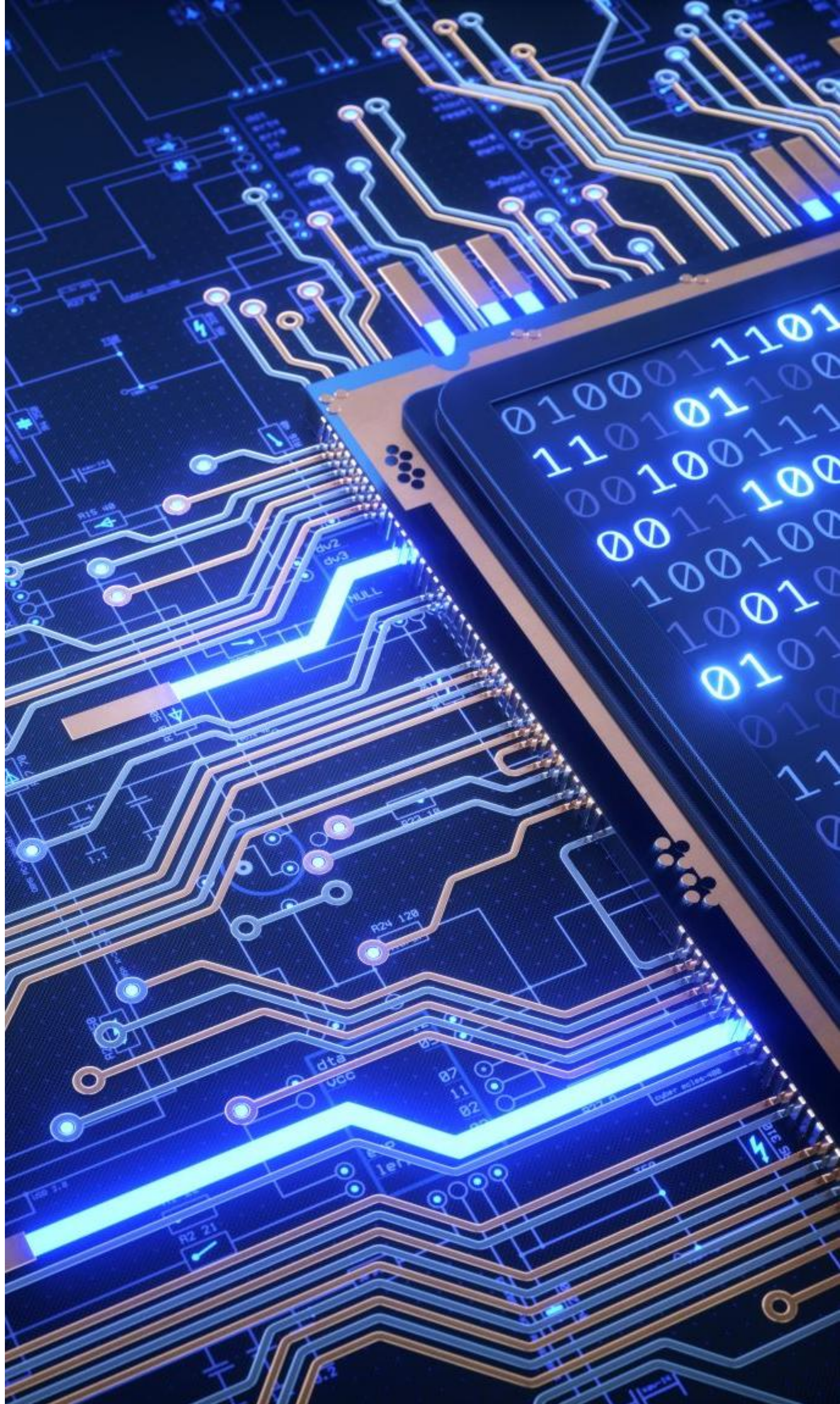

Module - 5
Design of Sequential Logic Circuits

Module Contents

- Latches
- Flipflops-SR, D, JK, T
- Buffer Registers
- Shift Registers- SISO, SIPO, PISO, PIPO
- Design of synchronous sequential circuit: state table and state diagrams
- Design of counters: Modulo-n, Johnson, Ring, Up/Down, Asynchronous counter
- Modeling of Sequential logic circuits using Verilog HDL.

Register

- A register is a group of flip-flops, each one of which shares a common clock and is capable of storing one bit of information.
- An **n-bit register** consists of a group of **n flip-flops** capable of storing **n bits** of binary information.
- In addition to the flip-flops, a register may have combinational gates that perform certain data-processing tasks.
- Various types of registers are available commercially. The simplest register is one that consists of only flip-flops, without any gates.

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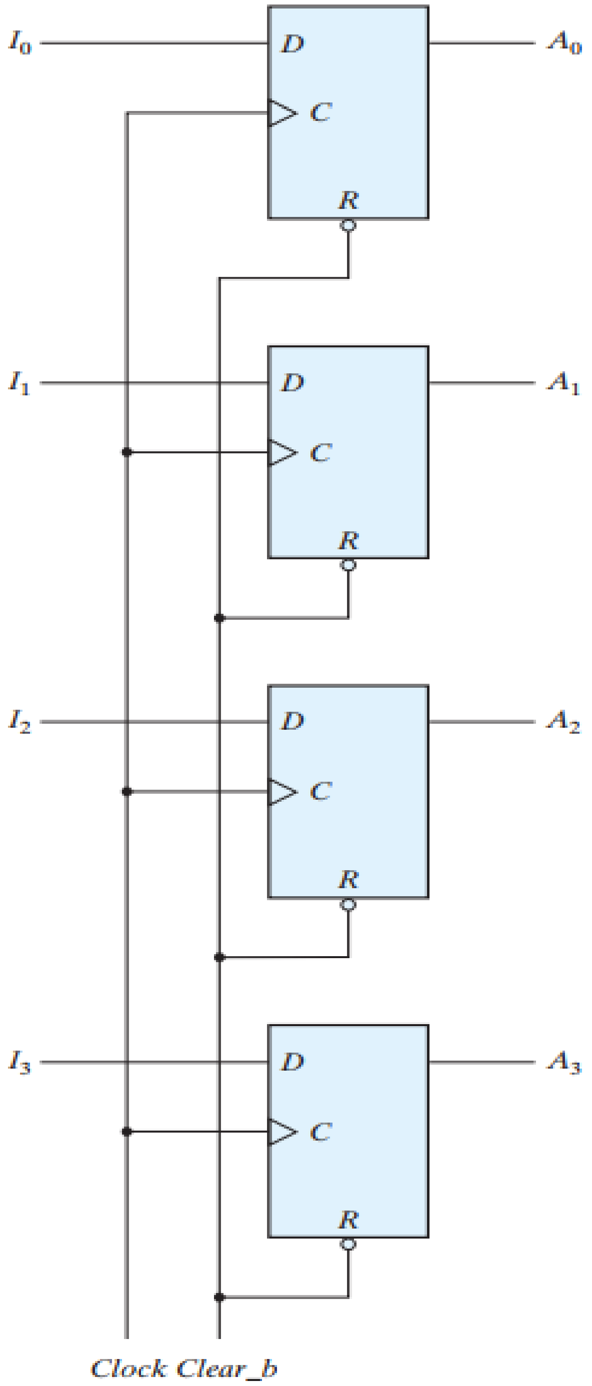
Four-bit register



- The common clock input triggers all flip-flops on the positive edge of each pulse, and the binary data available at the four inputs are transferred into the register.
- The value of (I3, I2, I1, I0) immediately before the clock edge determines the value of (A3,A2,A1,A0) after the clock edge.
- The four outputs can be sampled at any time to obtain the binary information stored in the register.
- The input Clear_b goes to the active-low R (reset) input of all four flip-flops.
- When this input goes to 0, all flip-flops are reset asynchronously.
- The Clear_b input is useful for clearing the register to all 0’s prior to its clocked operation.

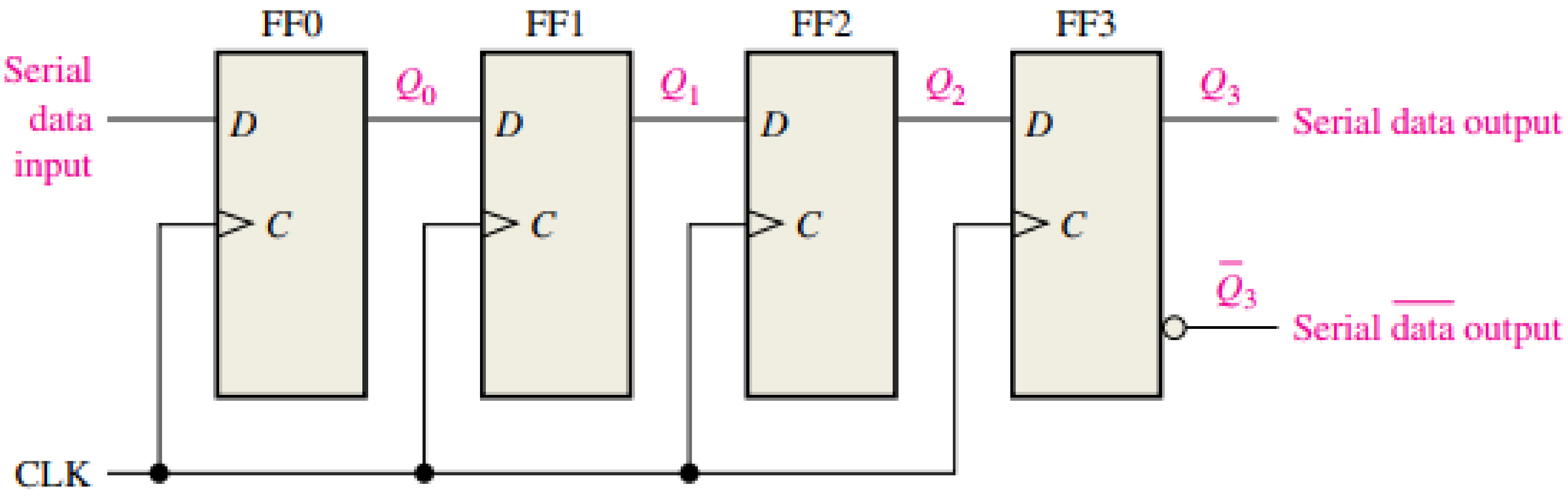
Shift Register

- A register capable of shifting the binary information held in each cell to its neighboring cell, in a selected direction, is called a shift register.
- All flip-flops receive common clock pulses, which activate the shift of data from one stage to the next.

Types of Shift Registers

- **Serial-in to Serial-out (SISO)** - the data is shifted serially "IN" and "OUT" of the register, one bit at a time in either a left or right direction under clock control.
- **Serial-in to Parallel-out (SIPO)** - the register is loaded with serial data, one bit at a time, with the stored data being available at the output in parallel form.
- **Parallel-in to Serial-out (PISO)** - the parallel data is loaded into the register simultaneously and is shifted out of the register serially one bit at a time under clock control.
- **Parallel-in to Parallel-out (PIPO)** - the parallel data is loaded simultaneously into the register and transferred together to their respective outputs by the same clock pulse.

Serial-in to Serial-out (SISO)



Input four bits 1010

CLK	FF0 (Q_0)	FF1 (Q_1)	FF2 (Q_2)	FF3 (Q_3)
Initial	0	0	0	0
1	**0**	0	0	0
2	**1**	**0**	0	0
3	**0**	**1**	**0**	0
4	**1**	**0**	**1**	**0**

Operation

- Input four bits, 1010, into the shift register, beginning with the least significant bit. (The register is initially clear)
- The 0 is put onto the serial data input line, making the data bit $D = 0$ for FF0. When the **first clock pulse** is applied, FF0 is reset, thus storing the 0.
- Next the second bit, which is a 1, is applied to the serial data input, making D = 1 for FF0 and D = 0 for FF1 because the D input of FF1 is connected to the Q0 output. When **the second clock pulse** occurs, the 1 on the data input is shifted into FF0, causing FF0 to set; and the 0 that was in FF0 is shifted into FF1.
- The third bit, a 0, is now put onto the serial data input line, and a **third clock pulse** is applied. The 0 is entered into FF0, the 1 stored in FF0 is shifted into FF1, and the 0 stored in FF1 is shifted into FF2.
- The last bit, a 1, is now applied to the serial data input, and a **fourth clock pulse** is applied. This time the 1 is entered into FF0, the 0 stored in FF0 is shifted into FF1, the 1 stored in FF1 is shifted into FF2, and the 0 stored in FF2 is shifted into FF3. This completes the serial entry of the four bits into the shift register, where they can be stored for any length of time as long as the flip-flops have dc power.

CLK	FF0 (Q_0)	FF1 (Q_1)	FF2 (Q_2)	FF3 (Q_3)
Initial	0	0	0	0
1	**0**	0	0	0
2	**1**	**0**	0	0
3	**0**	**1**	**0**	0
4	**1**	**0**	**1**	**0**

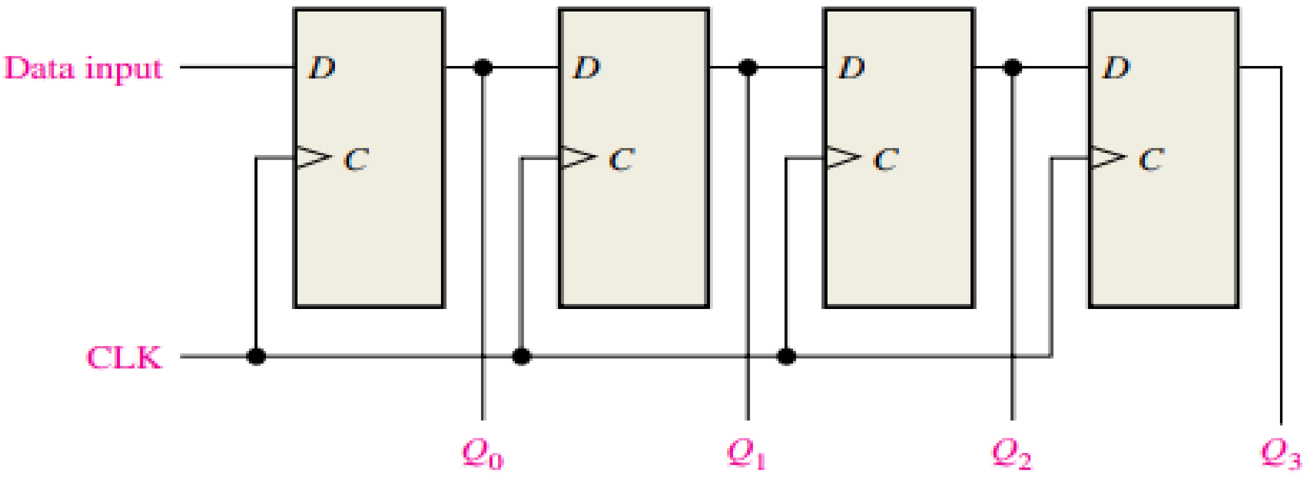
The serial entry of the four bits into the shift register requires four clock pulses.

Contd…

CLK	FF0 (Q_0)	FF1 (Q_1)	FF2 (Q_2)	FF3 (Q_3)
Initial	1	0	1	0
5	0	1	0	1
6	0	0	1	0
7	0	0	0	1
8	0	0	0	0

- If you want to get the data out of the register, the bits must be shifted out serially.
- To shift all the four bits out of the shift register, again four clock pulsed needed.

Serial-in to Parallel-out (SIPO)



CLK	FF0 (Q_0)	FF1 (Q_1)	FF2 (Q_2)	FF3 (Q_3)
Initial	0	0	0	0
1	**0**	0	0	0
2	**1**	**0**	0	0
3	**0**	**1**	**0**	0
4	**1**	**0**	**1**	**0**

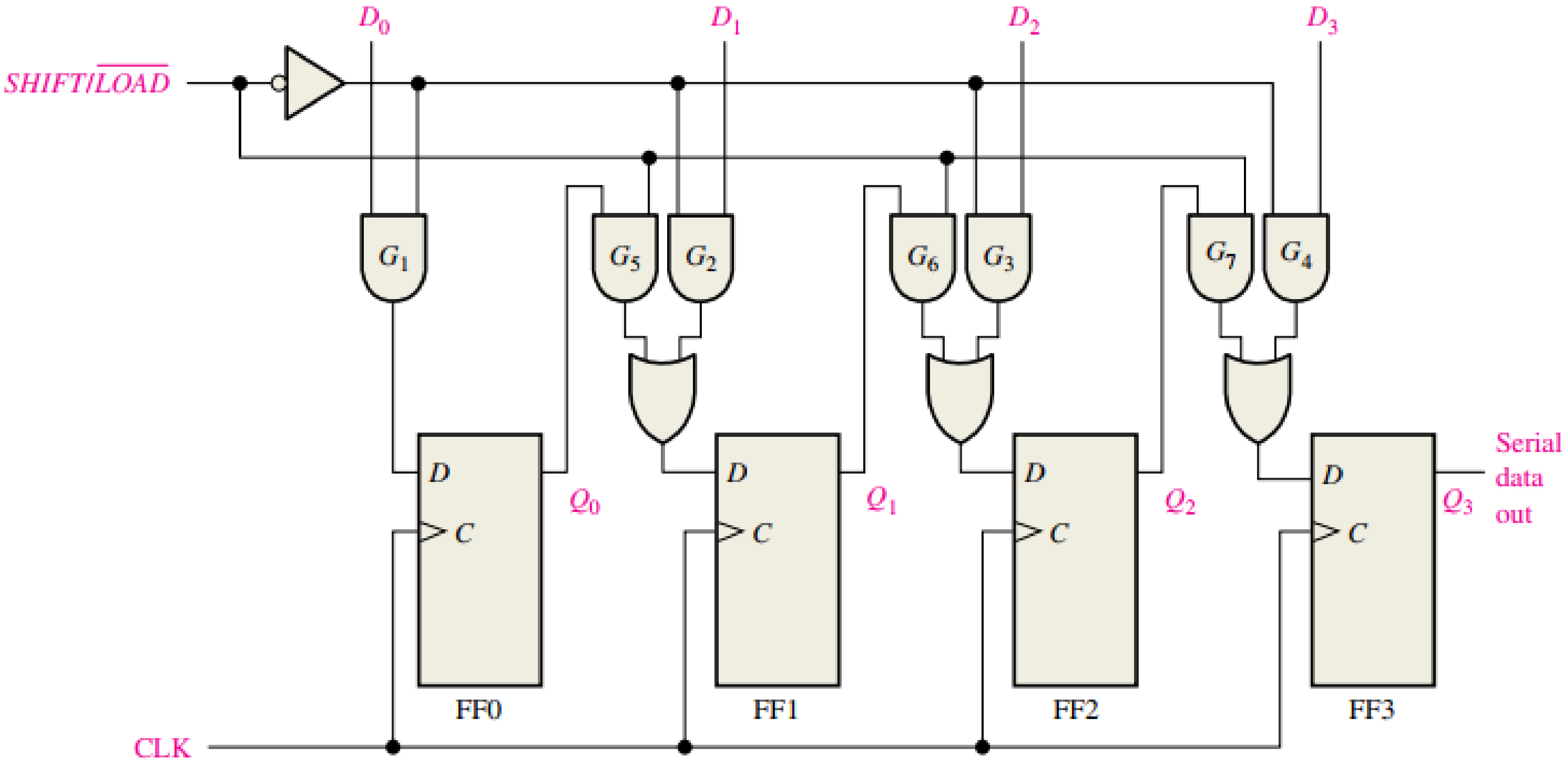
- Data bits are entered serially (least-significant bit first) into a serial in/parallel out shift register in the same manner as in serial in/serial out shift registers.
- The difference is the way in which the data bits are taken out of the register; in the parallel out register, the output of each stage is available.
- Once the data bits are stored, each bit appears on its respective output line, and all bits are available simultaneously, rather than on a bit-by-bit basis as with the serial output.

✓ The serial entry of the four bits into the shift register requires four clock pulses.

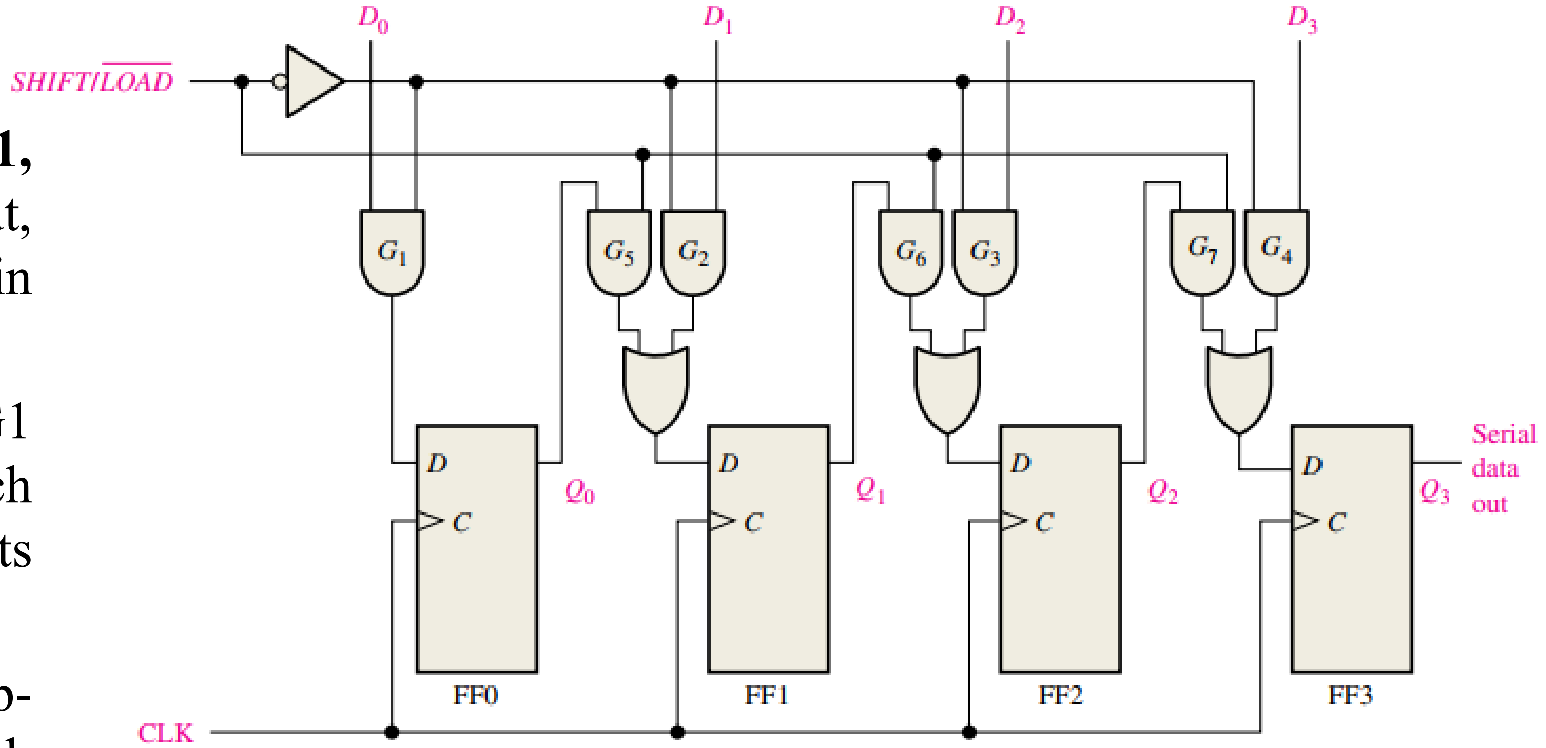
✓ After four clock pulses the input data is available at the output of shift register.

Parallel-in to Serial-out (PISO)

- For a register with parallel data inputs, the bits are entered simultaneously into their respective stages on parallel lines rather than on a bit-by-bit basis on one line as with serial data inputs. The serial output is the same as in serial in/serial out shift registers, once the data are completely stored in the register.

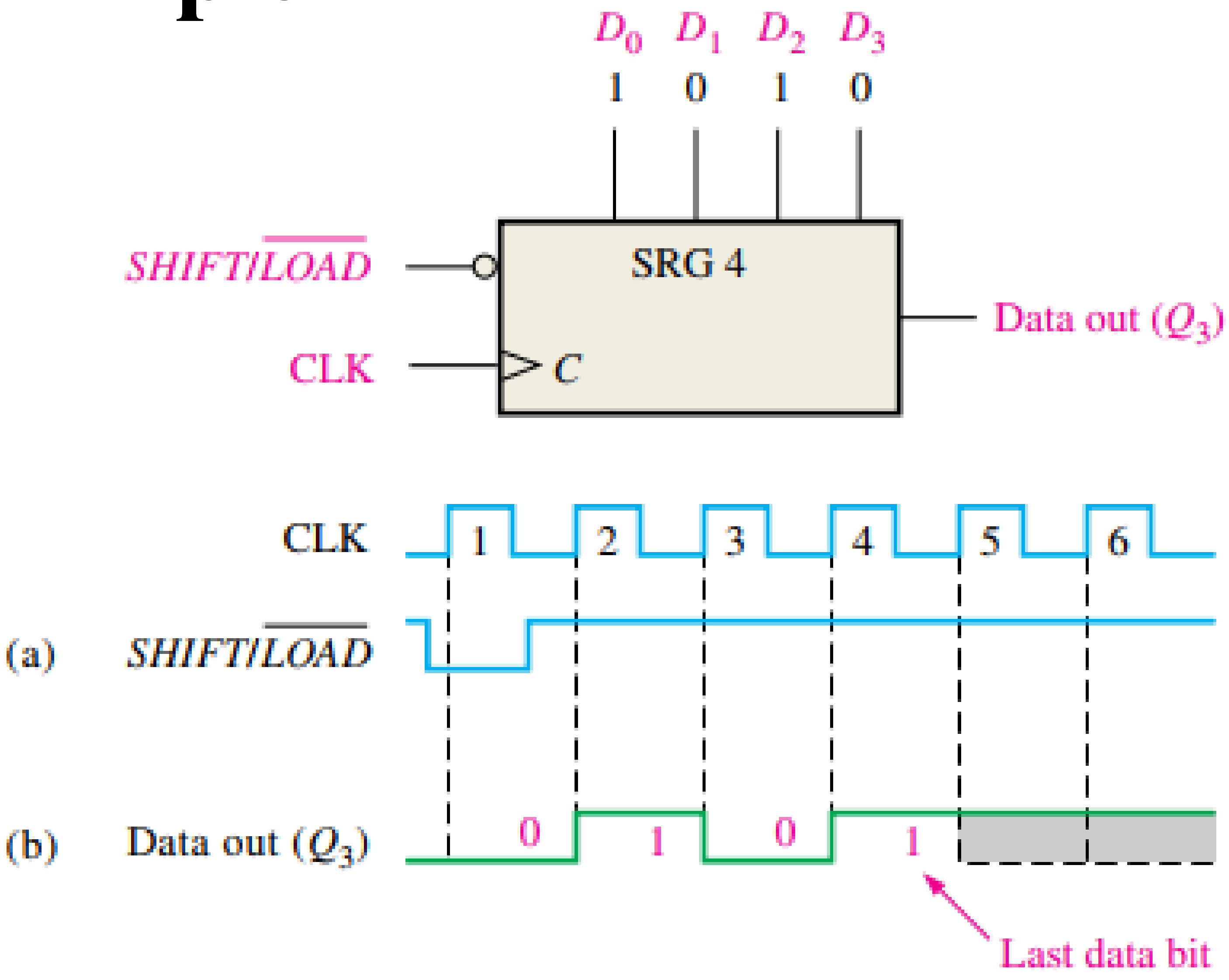


Operation



- There are four data-input lines, **D0, D1, D2**, and **D3**, and a $\boldsymbol{SHIFT/\overline{LOAD}}$ input, which allows four bits of data to **load** in parallel into the register.
- When $\boldsymbol{SHIFT/\overline{LOAD}}$ is **LOW**, gates G1 through G4 are enabled, allowing each data bit to be applied to the D input of its respective flip-flop.
- When a **clock pulse** is applied, the flip-flops with D = 1 will set and those with D = 0 will reset, thereby storing all four bits simultaneously.
- When $\boldsymbol{SHIFT/\overline{LOAD}}$ is **HIGH**, gates G1 through G4 are disabled and gates G5 through G7 are enabled, allowing the data bits to shift right from one stage to the next.
- The OR gates allow either the normal shifting operation or the parallel data-entry operation, depending on which AND gates are enabled by the level on the $\boldsymbol{SHIFT/\overline{LOAD}}$ input.
- Notice that FF0 has a single AND to disable the parallel input, D0. It does not require an AND/OR arrangement because there is no serial data in.

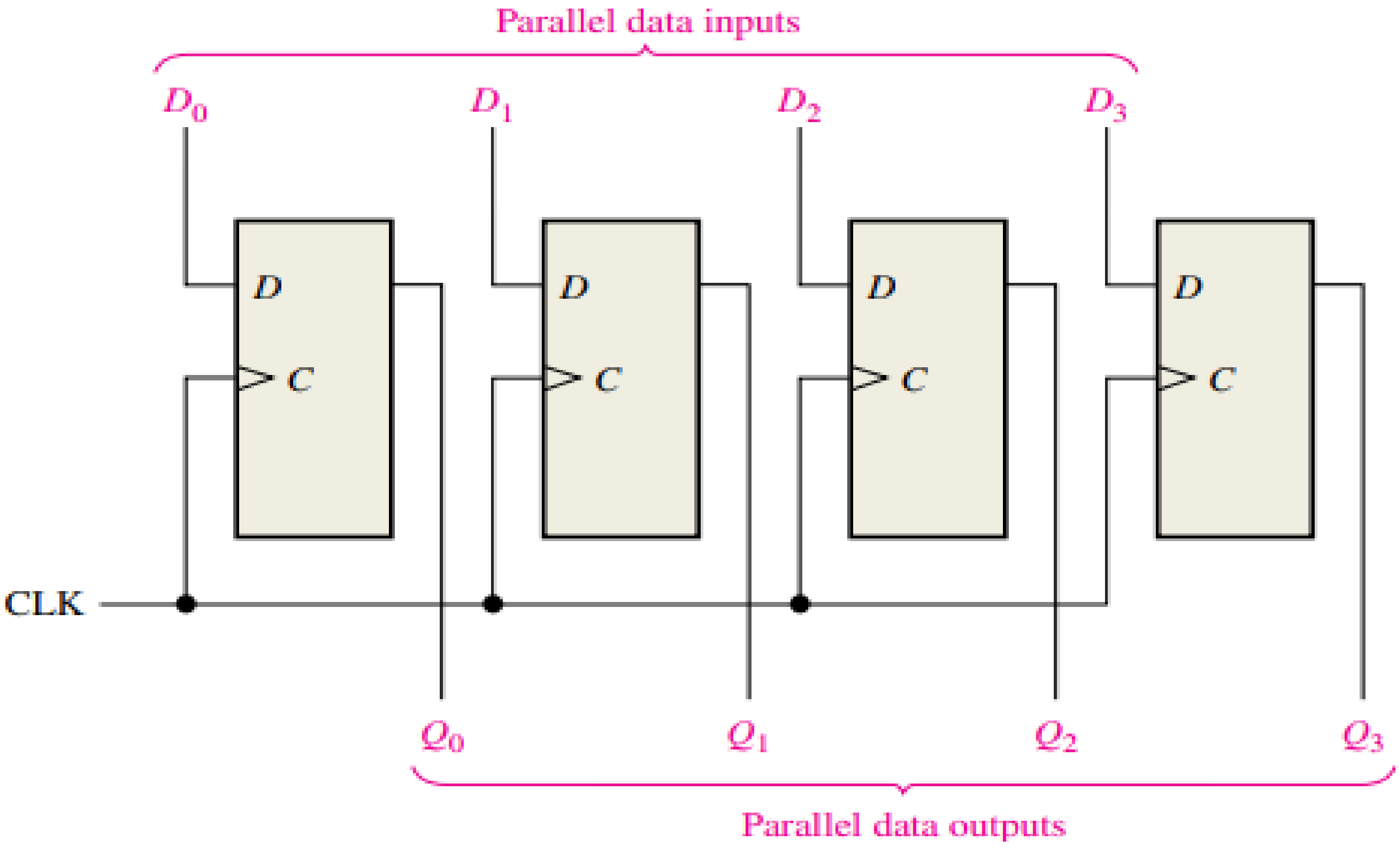
Example



D_0 D_1 D_2 D_3
1 0 1 0
SRG 4
$SHIFT/\overline{LOAD}$
CLK
C
Data out (Q_3)
CLK
1 2 3 4 5 6
(a) $SHIFT/\overline{LOAD}$
(b) Data out (Q_3)
0 1 0 1
Last data bit

Parallel-in to Parallel-out (PIPO)

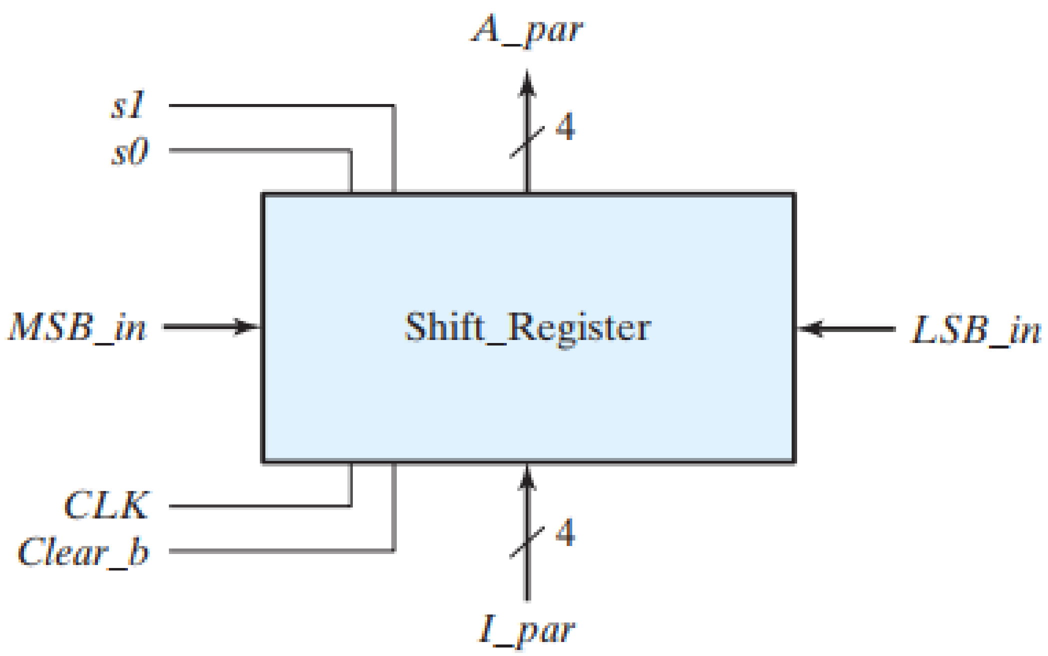
- Both parallel entry and parallel output of data have already been discussed.
- The parallel in/parallel out register employs both methods.
- Immediately following the simultaneous entry of all data bits, the bits appear on the parallel outputs.



Universal Shift Register

- **The most general shift register has the following capabilities**:
  - ❖A *clear* control to clear the register to 0.
  - ❖A *clock* input to synchronize the operations.
  - ❖A *shift-right* control to enable the shift-right operation and the *serial input* and *output* lines associated with the shift right.
  - ❖A *shift-left* control to enable the shift-left operation and the *serial input* and *output* lines associated with the shift left.
  - ❖A *parallel-load* control to enable a parallel transfer and the *n* input lines associated with the parallel transfer.
  - ❖*n* parallel output lines.
  - ❖A control state that leaves the information in the register unchanged in response to the clock.
- A register capable of shifting in one direction only is a ***unidirectional*** **shift register**.
- One that can shift in both directions is a ***bidirectional*** **shift register**.
- If the register has **both shifts and parallel-load capabilities**, it is referred to as a ***universal shift register***.

Universal Shift Register



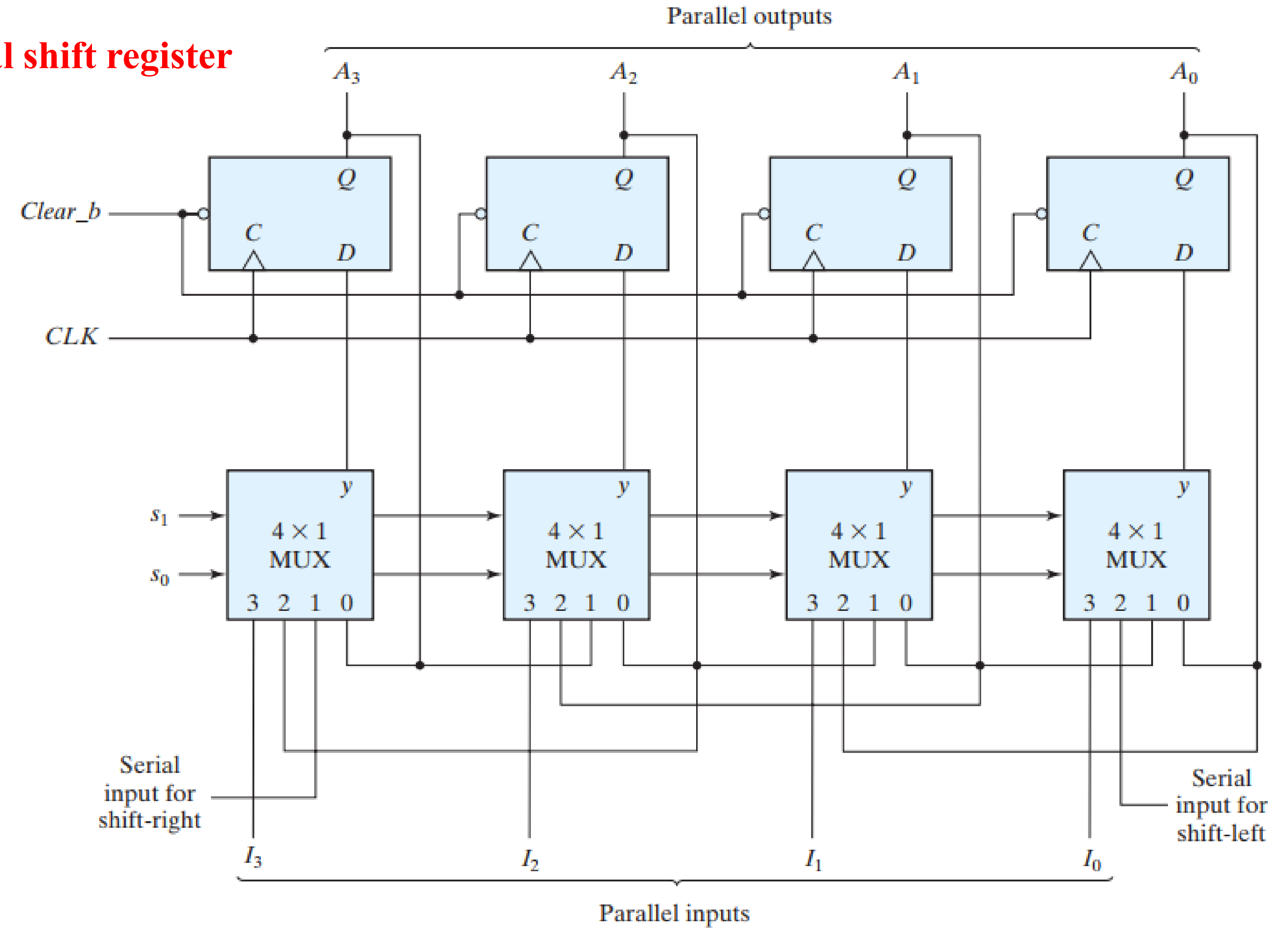
Four-bit universal shift register

Mode Control		
s_1	s_0	**Register Operation**
0	0	No change
0	1	Shift right
1	0	Shift left
1	1	Parallel load

- The circuit consists of four D flip-flops and four multiplexers.
- The **four multiplexers have two common selection inputs s1 and s0**.
- Input 0 in each multiplexer is selected when s1s0 = 00, input 1 is selected when s1s0 = 01, and similarly for the other two inputs.
- The selection inputs control the mode of operation of the register according to the function entries.

Universal Shift Register

Four-bit universal shift register



Operation

- When **s1s0 = 00**, the present value of the register is applied to the D inputs of the flip-flops. **This condition forms a path from the output of each flip-flop into the input of the same flip-flop**, so that the output recirculates to the input in this mode of operation. The next clock edge transfers into each flip-flop the binary value it held previously, and no change of state occurs.
- When **s1s0 = 01**, terminal 1 of the multiplexer inputs has a path to the D inputs of the flip-flops. This causes a shift-right operation, with the serial input transferred into flip-flop A3.
- When **s1s0 = 10**, a shift-left operation results, with the other serial input going into flip-flop A0.
- Finally, when **s1s0 = 11**, the binary information on the parallel input lines is transferred into the register simultaneously during the next clock edge.

Note:

Data enters MSB_in for a shift-right operation and enters LSB_in for a shift-left operation.

Clear_b is an active-low signal that clears all of the flip-flops.

Any Queries!

Thank you!